

Section 1: Metadata

to be filled by the student

1.1. Project Information to be filled by the student

Title: Restaurant Management System	
Section:	Instructor:

1.2. Student(s) Information

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Name:	ID:
Section:	Batch:

Submission guideline: Save your project proposal as a pdf file and rename as Project Proposal_L1_ProposedTitle where L1 is to be replaced with your section

Section 2: The Project

to be filled by the student

2.1. Project Description: *Please provide a brief introduction of the project including its scope.*

Introduction:

Our project is centered around designing and implementing a Database Management System for a restaurant. This system aims to streamline restaurant operations by efficiently managing various key elements like menu items, customer orders, inventory, staff, and billing. The database will be used to store essential information, ensuring that data can be accessed, updated, and retrieved in real-time to enhance the overall management process. This system will support features such as order tracking, table reservations, staff management, and generating reports for sales and inventory, helping to optimize workflow and improve customer service.

Purpose and functionality:

For the customers this system offers services like menu browsing, table reservations, order placement and tracking, payments and feedbacks. This is aimed to enhance overall customer experience by simplifying the process through one central system.

For the restaurant staff the system aims to streamline the preparation and serving process in order to reduce waiting times even during rush hour, which has been a lasting problem in restaurants today. The services include order tracking and management, table assignment and reservation handling, and inventory monitoring.

Lastly for the admin staff (such as managers or IT support) the system includes functions such as staff management, menu and pricing management along with financial transactions and sales reports, and user role and access control.

Scope:

Our project covers a wide range of functions and dives deep into how a restaurant assisted by technology looks like all while providing features like data security and real-time updates along with a user-friendly interface to bring it all together. This system is designed with scalability in mind to accommodate for future expansions such as new branches. The system focuses on streamlining customer interactions, improving staff efficiency, and giving admins complete control over restaurant operations, all through a centralized, user-friendly system ensuring an organized workflow, better customer service, and enhanced decision-making capabilities for the business.

2.2 Functional Requirements

This section describes each function/feature provided by your system. These functions are logically grouped into modules based on their purposes. The users in your system must be categorized such as client, customer or administrator etc. These users will be accessing the database with the level of access that they are authorized with.

Sample functional Requirements:

Module 1: Registration

- Function 1: Register an account
 - Function 1a: This allows users such as waitstaff, managers, chefs, admin and customers to register to the platform. The registration form captures essential data such as name, email, username, password and user type.
 - Function 1b: The information is then added to the system database allowing role tracking.

Module 2: Menu Management

- Function 1: Items can be added to the menu with details like name, category, price and description.
- Function 2: Existing Menu items can be selected and edited such as changing prices.
- Function 3: Selected items can be deleted from the menu when they are no longer available.

Module 3: Order Management

- Function 1: Orders can be placed by either customers or waitstaff and are assigned order IDs which can then be used for tracking and preparation. The details include the selected menu items, table number and special requests if there are any.
- Function 2: Existing orders can be edited or cancelled by waitstaff if necessary.

Module 4: Table Reservation.

- Function 1: Customers or can make reservations for a specific date and time with additional details like party size. The reservations are stored in the database and can be viewed by the staff
- Function 2: Customers can update or cancel reservations through the system which will then be reflected in the system.

Module 5: Staff Management

- Function 1: Mangers can add new staff members with details like name, role, contact info and working shifts. Staff details can be updated or removed as per needed.
- Function 2: Managers can assign and manage staff shifts to ensure adequate coverage during operating hours (especially rush hour).

Module 6: Customer Feedback and Reviews

- Function 1: Customers can give feedback about their dining experience including ratings and comments about the food and service provided.
- Function 2: Managers can view customer feedback and ratings to assess service and food quality and address any issues.

Module 7: Inventory Management

- Function 1: The system tracks inventory levels and of ingredients and supplies used in the restaurant. Alerts are triggered when inventory levels are low
- Function 2: Staff can update inventory levels after receiving new shipments or at certain times during the day.

Module 8: Billing and Payment

- Function 1: The system generates bills for orders, including itemized details of the order and applicable taxes.
- Function 2: The system can process various payment methods such as cash, credit card and online payments.

2.3. Planned Schedule: *Kindly list the start/end dates and the timeline for the achievement of any intermediate milestones and the expected contribution to be made by the participant(s).*

Work starts on ERD	Week 4
Work starts on project	Week 5
Interim ERD	Week 8
ERD submission and interim demo	Week 13
Project finalization and submission	Week 15

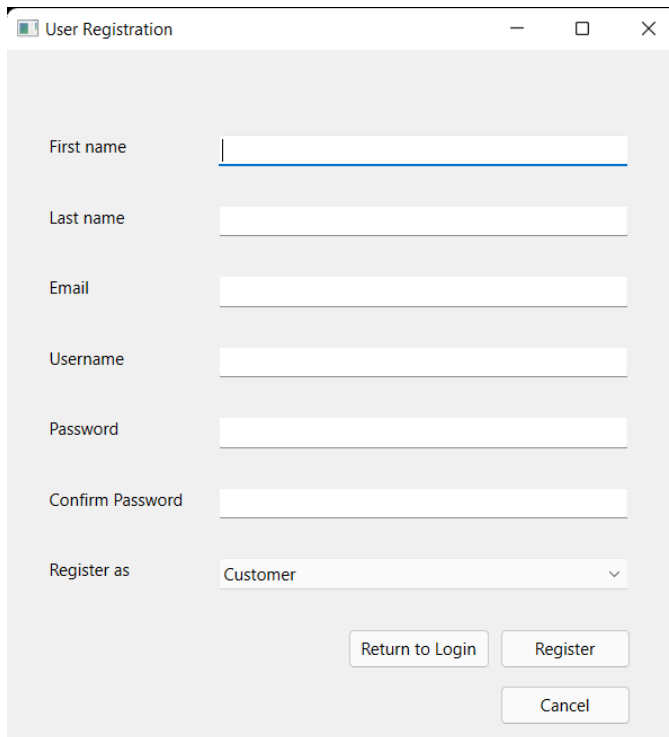
2.4. Technology Stack: *If you are utilizing any language or database besides PyQt and SQL Server, please complete this section; otherwise, leave it blank. Specify the programming language and database management system intended for constructing this application, as well as the application type (Desktop, Web, or Mobile).*

Main Programming Language: Python

Database Management System: SQL Management Studio

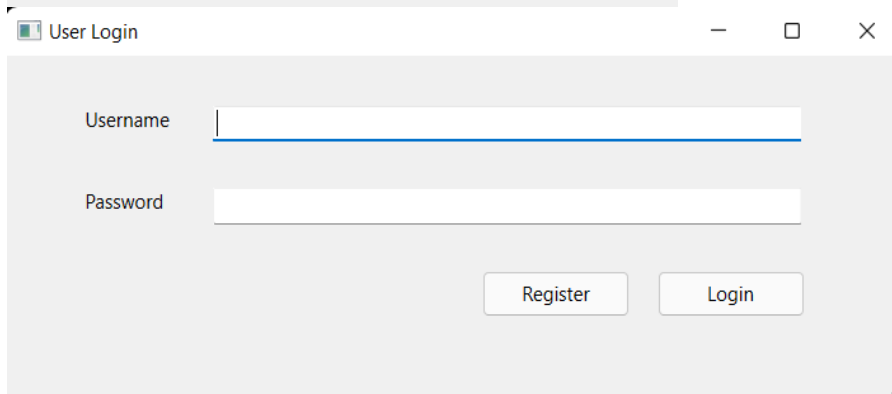
Application Type: Desktop

2.5. Screens: Provide images of all application screens, showcasing clear input and corresponding outputs. Ensure each image includes a concise caption explaining user action and expected/observed output. You can create these screens using Qt Designer.



A screenshot of a Qt Designer window titled "User Registration". The window contains a form with the following fields: "First name", "Last name", "Email", "Username", "Password", "Confirm Password", and a "Register as" dropdown menu currently set to "Customer". At the bottom of the form are three buttons: "Return to Login", "Register", and "Cancel".

These screens are the user login and registration screens. The user will login using their username and password after registration.



A screenshot of a Qt Designer window titled "User Login". The window contains a form with two fields: "Username" and "Password". At the bottom of the form are two buttons: "Register" and "Login".

CS355 – Database Project Proposal Form

Menu management

Item Name

Category

Price

Description

Menu

This screen is used to add items to the menu and the whole menu is also displayed in the bottom.

CS355 – Database Project Proposal Form

Order Management

Place an Order

Table Number

Menu Items

Special Requests

Place Order

Order Status

Order ID

View Status

Cancel Order

Order Status

Table Number

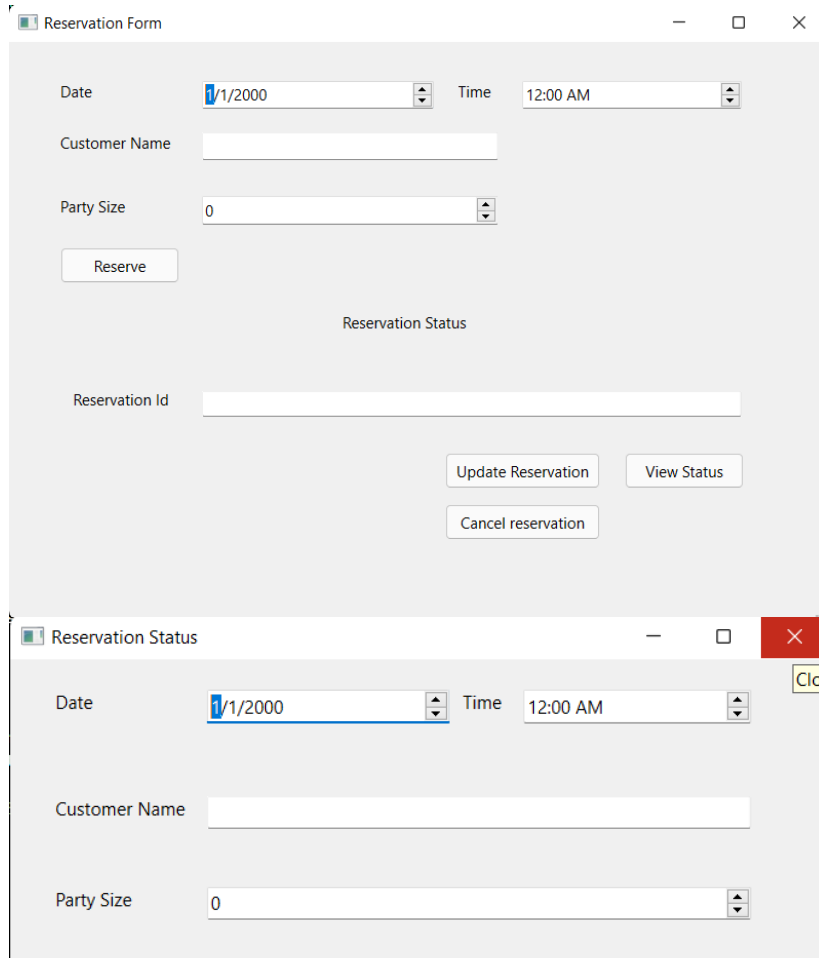
Item

Price

Special Requests

This screen is used to place and view an order from the top part and also view the status of the order using the order ID in the bottom part.

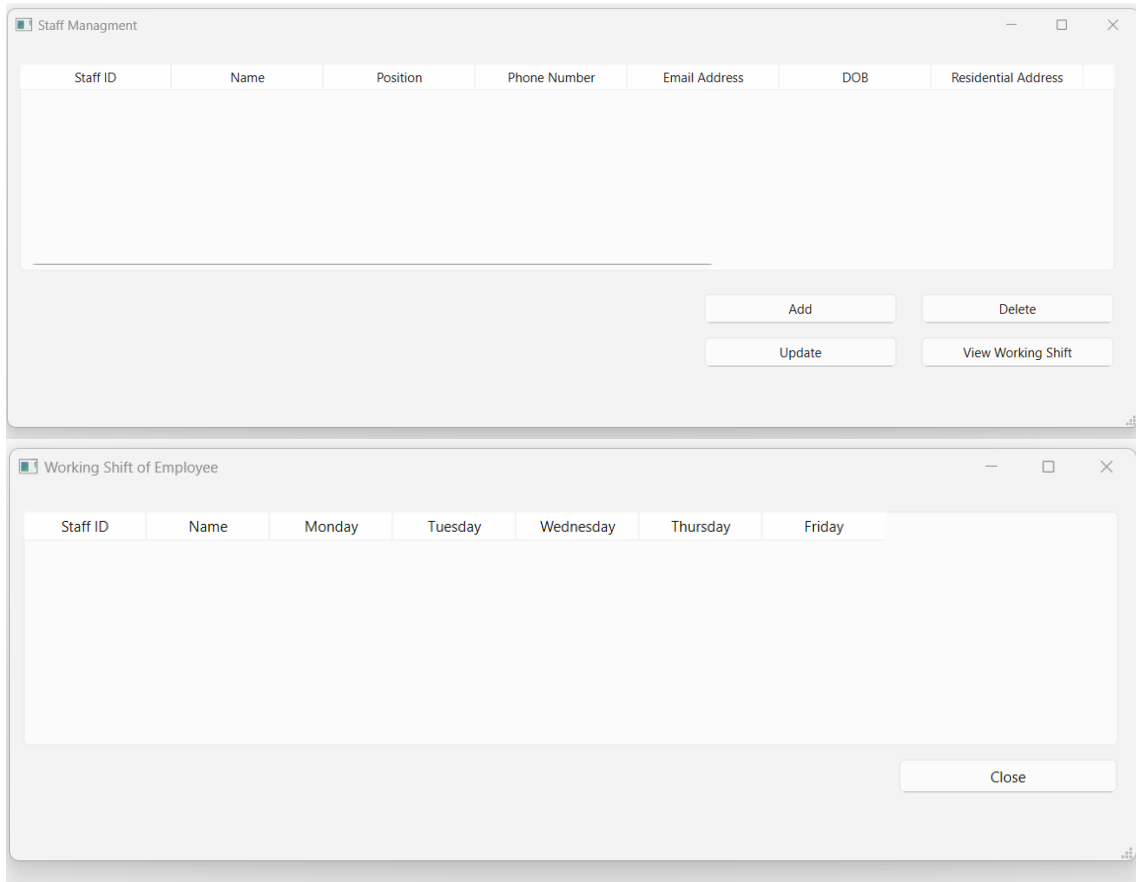
CS355 – Database Project Proposal Form



The image shows two overlapping application windows. The top window, titled "Reservation Form", contains fields for "Date" (set to 1/1/2000), "Time" (set to 12:00 AM), "Customer Name", and "Party Size" (set to 0). It features a "Reserve" button and a "Reservation Status" section with a "Reservation Id" field and buttons for "Update Reservation", "View Status", and "Cancel reservation". The bottom window, titled "Reservation Status", displays the same "Date", "Time", "Customer Name", and "Party Size" fields. A "Close" button is visible in the top right corner of this window.

This screen is used to make and view table reservations using the reservation ID. The bottom screen is for viewing only.

CS355 – Database Project Proposal Form



The image displays two windows from a database application. The top window, titled "Staff Management", features a table with columns: Staff ID, Name, Position, Phone Number, Email Address, DOB, and Residential Address. Below the table are four buttons: "Add", "Delete", "Update", and "View Working Shift". The bottom window, titled "Working Shift of Employee", features a table with columns: Staff ID, Name, Monday, Tuesday, Wednesday, Thursday, and Friday. A "Close" button is located at the bottom right of this window.

Staff ID	Name	Position	Phone Number	Email Address	DOB	Residential Address
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Add Delete
Update View Working Shift

Staff ID	Name	Monday	Tuesday	Wednesday	Thursday	Friday
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Close

These screens can be used to view and edit existing staff while also providing the functionality to add or remove people. The staff info is taken using the same user registration screen attached previously.

CS355 – Database Project Proposal Form

Customer Feedback

Resturant Customer Satisfaction Form

Please rate your satisfaction with our resturant based on your visit here

Order Number:

Food Quality

Menu Diversity
☐ Very Poor
☐ Poor
☐ Fair
☐ Good
☐ Excellent

Freshness
☐ Very Poor
☐ Poor
☐ Fair
☐ Good
☐ Excellent

Service Quality

Response Time of The Staff
☐ Very Poor
☐ Poor
☐ Fair
☐ Good
☐ Excellent

Politeness of the Staff
☐ Very Poor
☐ Poor
☐ Fair
☐ Good
☐ Excellent

Submit

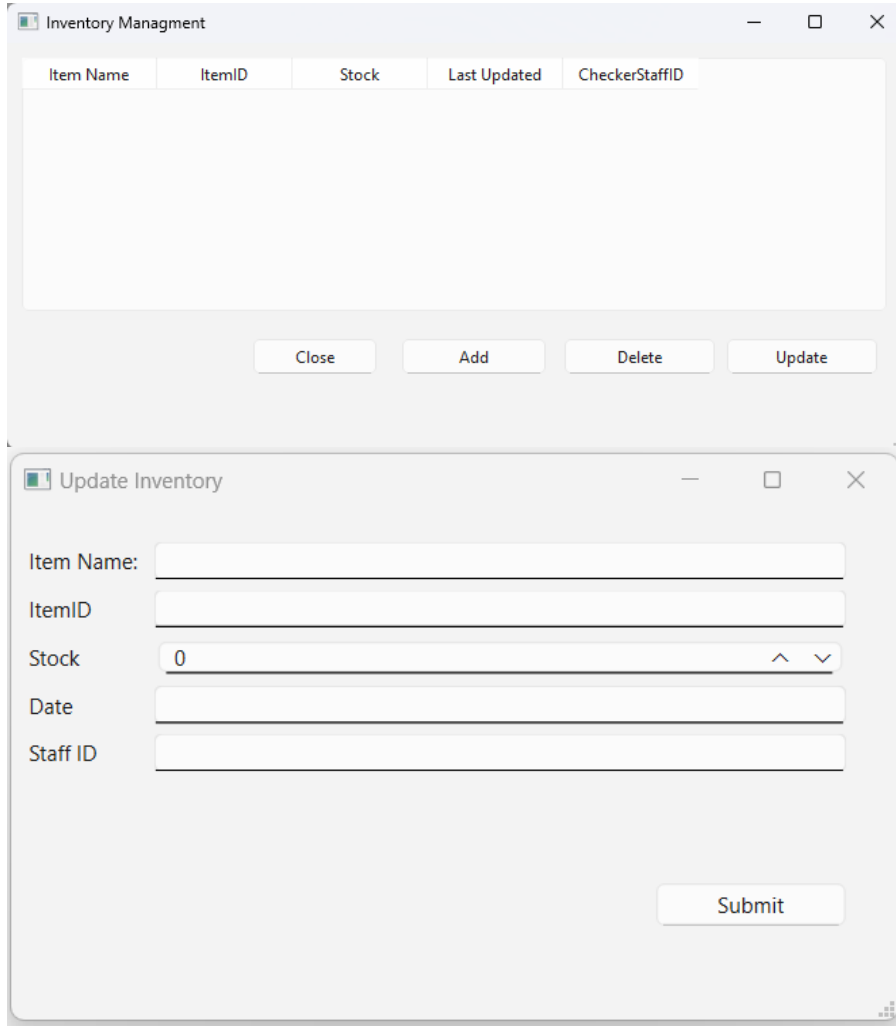
Customers Feedback View

OrderID	Freshness	Menu Diversity	Response Time of The Staff

Close

These screens can be used to receive and view customer feedback.

CS355 – Database Project Proposal Form



The image shows two overlapping windows from a database application. The top window, titled "Inventory Managment", features a table with five columns: "Item Name", "ItemID", "Stock", "Last Updated", and "CheckerStaffID". Below the table are four buttons: "Close", "Add", "Delete", and "Update". The bottom window, titled "Update Inventory", contains five input fields labeled "Item Name:", "ItemID", "Stock", "Date", and "Staff ID". The "Stock" field is a spinner control showing the value "0". A "Submit" button is located at the bottom right of this window.

This screen is used to view items in the inventory and also have the functionality to update, add and, delete items. The bottom screen is used for both updating and adding new items to the inventory.

Bill Print and Order Complete

Order ID

Date

1/1/2000 12:00 AM

^ v

Menu Items

Payment Type

v

Voucher Code

Served By:

Subtotal

Tax

Total

Payment Received

Print

Billing and Payment

Order ID

StaffID

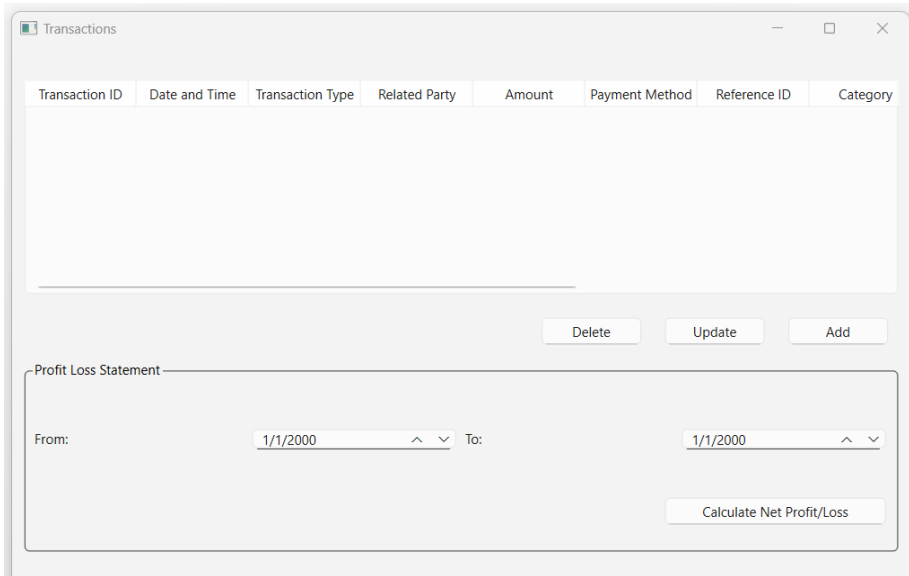
Table Number

Payment Type

Generate Bill

These screens are used to generate bill of the selected orders from the bottom screen. The top screen then displays all the details and has the functionality to generate the bill.

CS355 – Database Project Proposal Form



The screenshot shows a window titled "Transactions" with a table header and a "Profit Loss Statement" section.

Transaction ID	Date and Time	Transaction Type	Related Party	Amount	Payment Method	Reference ID	Category
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Buttons: Delete, Update, Add

Profit Loss Statement

From: 1/1/2000 To: 1/1/2000

Calculate Net Profit/Loss

This screen is used to view all the transactions and has the functionality to update or delete selected transactions. The bottom part can be used to generate a report based on the start and end dates provided.